Introduction

Data visualization is widely used to communicate information across many fields, including business, research, and public policy. Data graphics are powerful tools for reasoning and communicating quantitative information (Tufte, 2001). However, many people lack the skills needed to accurately interpret different types of charts, which can lead to misunderstanding or incorrect conclusions. As visualization tools become more accessible, it is increasingly important to improve data visualization literacy.

This study examines how different visualization types affect accuracy and response time when interpreting categorical data. By comparing multiple chart types, we aim to determine which visualizations are most effective for supporting accurate and efficient data interpretation. The research question guiding this study is: ***How does visualization type affect how quickly and accurately people answer questions about categorical data?***

Understanding these differences can help guide better design choices in data visualization.

Method

Participants completed a series of tasks using four visualization types: bar charts, line graphs, treemaps, and stacked charts. For each visualization, participants answered questions related to categorical data while their accuracy and response time were recorded. After each task, participants reported their confidence, and at the end, they indicated their preferred visualization type. The data were analyzed by calculating average accuracy and response time for each visualization type, and a repeated-measures ANOVA was used to assess statistical significance.

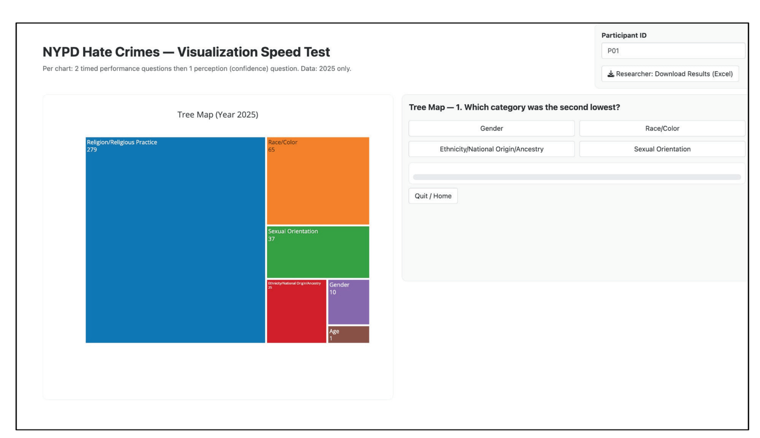

Fig. 1. Example of the task interface showing a treemap visualization and question presented to participants.

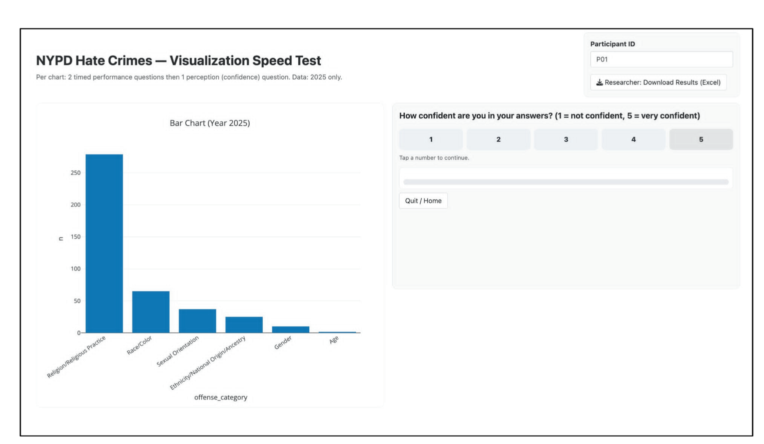

Fig. 2. Example of the confidence rating interface where participants reported how confident they were in their answer

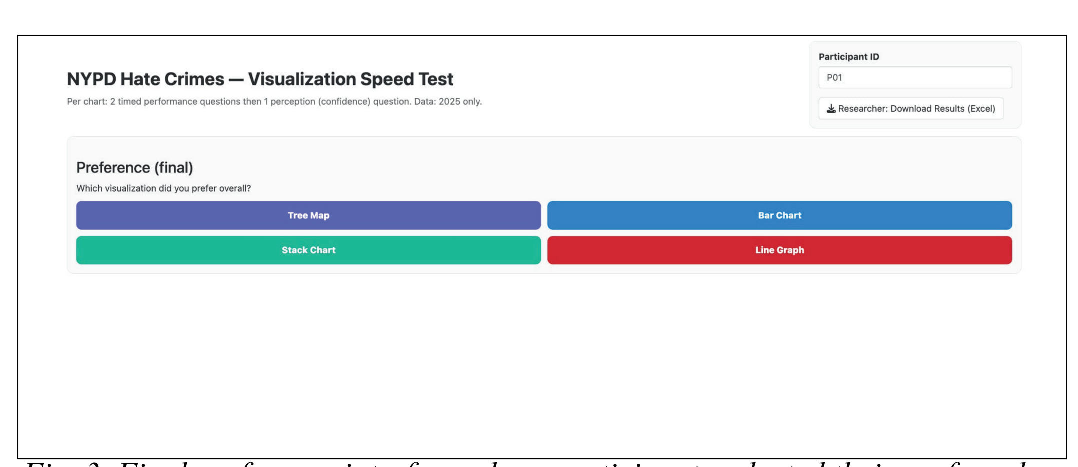

Fig. 3. Final preference interface where participants selected their preferred visualization type after completing all tasks

Results

Bar charts achieved the highest accuracy, while stack charts showed the lowest accuracy. Line graphs resulted in the fastest response times, with only small differences compared to bar charts in response time. A repeated-measures ANOVA indicated that visualization type had a statistically significant effect on performance ($p < 0.05$). These results suggest that visualization type influences both accuracy and efficiency when interpreting categorical data.

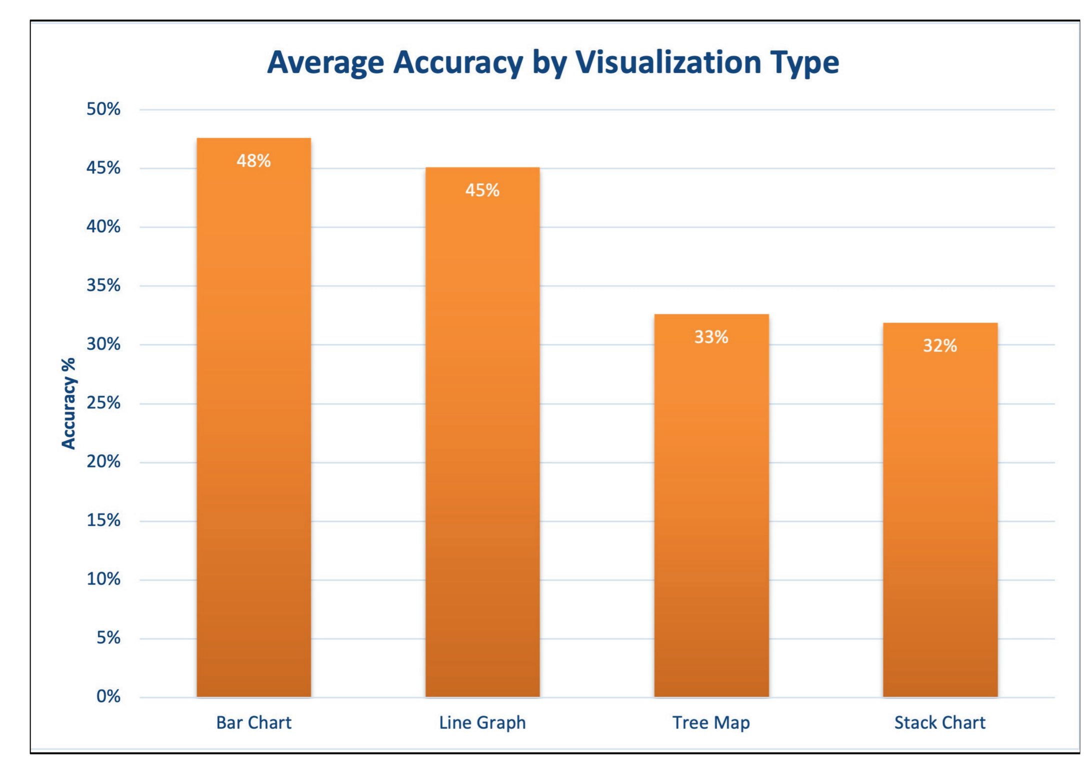

Fig. 4. Average accuracy across visualization types. Bar charts show the highest accuracy, while treemaps and stack charts have lower accuracy.

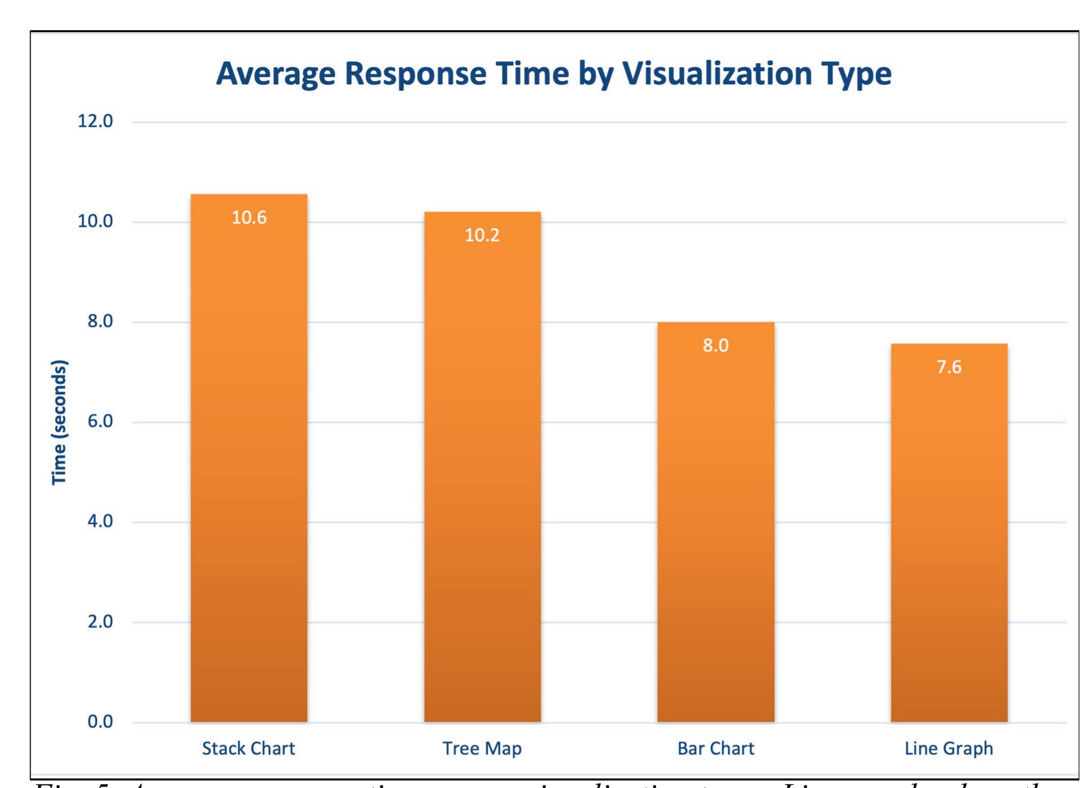

Fig. 5. Average response time across visualization types. Line graphs show the fastest response time, while treemaps and stack charts are slower.

Conclusion

This study demonstrates that visualization type significantly affects both the speed and accuracy of responses when interpreting categorical data. Bar charts were the most effective overall, achieving the highest accuracy with relatively fast response times, while stack charts consistently performed the worst in both measures. Although line graphs enabled slightly faster responses, the differences in speed were minimal compared to differences in accuracy. These findings suggest that selecting appropriate visualization types is critical for supporting accurate and efficient data interpretation, highlighting the important role of visualization design in decision-making (Imarticus Learning, 2025).

References

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